SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I, Shrimathi V_192571078 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Problem Statement:

A hospital is designing a **Rule-Based Logical Agent** for preliminary patient diagnosis. The knowledge base contains the following rules:

1. If a patient has **a Fever AND Cough**, then the patient has **a Possible Flu**.
2. If a patient has **Fever AND Rash**, then the patient has **Possible Measles**.
3. If a patient has **Cough AND Breathlessness**, then the patient has **Possible Pneumonia**.

Facts about **Patient A**:

- Fever = True
- Cough = True
- Breathlessness = True

The objectives are:

- (a) Represent the rules and facts using Propositional Logic.
- (b) Apply Forward Chaining to derive all possible diagnoses.
- (c) Apply Backward Chaining to verify whether Patient A has Pneumonia.

(a) Representation Using Propositional Logic

Definition of Symbols

Symbols	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
FL	Patient has Possible Flu
M	Patient has Possible Measles
P	Patient has Possible Pneumonia

Knowledge Base (KB)

Rule 1

If a patient has Fever and Cough, then the patient has Possible Flu.

$F \land C \rightarrow FL$

Rule 2

If a patient has Fever and Rash, then the patient has Possible Measles.

$F \land R \rightarrow M$

Rule 3

If a patient has Cough and Breathlessness, then the patient has Possible Pneumonia.

$C \land B \rightarrow P$

Facts

Patient A has:

F=True C=True B=True

or simply,

F,C,B

Knowledge Base Summary

Rule No	Logical Representation
R1	$F \land C \rightarrow FL$
R2	$F \land R \rightarrow M$
R3	$C \land B \rightarrow P$
Facts	F,C,B

Analysis

The knowledge base stores medical rules in propositional logic, where symptoms are represented as propositions and diagnoses are represented as conclusions. Logical inference enables the system to derive possible diseases based on the available patient symptoms.

Patient A exhibits Fever, Cough, and Breathlessness. Since these symptoms satisfy the conditions of Rule 1 and Rule 3, the logical agent can infer multiple possible diagnoses. However, Rule 2 cannot be applied because the patient does not have Rash.

(b) Forward Chaining

Concept

Forward Chaining is a **data-driven inference technique**. It begins with the known facts and repeatedly applies inference rules until no new facts can be derived or the desired conclusion is reached.

Initial Facts

Fever (F)

Cough (C)

Breathlessness (B)

Inference Process

Step1: Known Facts

F

C

B

Step2: Check Rule 1

Rule: $F \land C \rightarrow FL$

Since,

F = True

C = True

Rule 1 fires.

New Fact derived: Possible Flu (FL)

Step3: Check rule 2

$F \land R \rightarrow M$

Although Fever is True,

Rash is absent.

Therefore,

Rule 2 **does not fire**.

No new fact is generated.

Step 4: Check Rule 3

Rule: $C \land B \rightarrow P$

Since,
C=True

B=True

Rule 3 Fires

New Fact Derived: Possible Pneumonia (P)

Forward chaining Table:

Step	Rule Applied	Facts Used	New Fact Derived
1	Initial Facts	Fever, Cough, Breathlessness	—
2	R1	Fever + Cough	Possible Flu
3	R2	Fever + Rash	Not Applicable
4	R3	Cough + Breathlessness	Possible Pneumonia

Final Diagnoses

The Forward Chaining process derives the following diagnoses:

- **Possible Flu**
- **Possible Pneumonia**

No conclusion regarding **Measles** can be drawn because the symptom **Rash** is not present.

Analysis of Forward Chaining

Forward Chaining successfully identifies all possible diagnoses by examining every rule whose conditions match the available facts. Since it begins with observed symptoms and derives conclusions, it is particularly suitable for medical diagnosis systems where patient information is collected first and diseases are inferred afterward.

(c) Backward Chaining

Concept

Backward Chaining is a **goal-driven inference technique**. It starts with the desired conclusion (goal) and works backward to determine whether the available facts can satisfy the required conditions.

Goal: Possible Pneumonia (P)

Step 1

Search the Knowledge Base for a rule that concludes:

P

Rule Found:

$C \land B \rightarrow P$

Step 2

Break the goal into sub-goals.

Required:

Cough (C)

Breathlessness (B)

Step 3

Verify each sub-goal.

Knowledge Base contains:

C = True

B = True

Both conditions are satisfied.

Step 4

Since both sub-goals are true,

Rule 3 succeeds.

Therefore,

Possible Pneumonia = True

Goal Reduction Tree

```
      Possible Pneumonia (P)
             |
       Rule 3: C ∧ B → P
          /        \
         /          \
    Cough (C)    Breathlessness (B)
        |             |
      TRUE          TRUE
        \           /
         \         /
       Goal Proven ✓
```

Analysis of Backward Chaining

Backward Chaining efficiently verifies a specific diagnosis by starting with the target goal and checking only the rules relevant to that goal. Unlike Forward Chaining, which derives all possible conclusions, Backward Chaining minimizes unnecessary reasoning and is particularly useful in expert systems that need to confirm a suspected disease.

Engineering Interpretation

The Smart Medical Diagnosis System demonstrates how Artificial Intelligence can support healthcare professionals by applying logical reasoning techniques to patient symptoms. Knowledge Representation using propositional logic allows medical expertise to be stored in a structured form, while inference mechanisms such as Forward Chaining and Backward Chaining enable the system to derive and verify diagnoses systematically. Such systems assist doctors in preliminary diagnosis, improve decision-making speed, and reduce the possibility of overlooking important symptom combinations. Although the final diagnosis should always be confirmed by a medical professional, logical agents provide an effective decision-support tool in modern healthcare environments.

Critical Analysis

Advantages

- Provides consistent and systematic reasoning.

- Reduces human error in preliminary diagnosis.
- Supports quick identification of possible diseases.
- Easy to update by adding new medical rules.
- Efficiently handles multiple symptoms and diagnoses.

Limitations

- Limited to the rules stored in the knowledge base.
- Cannot diagnose diseases outside the predefined rules.
- Does not account for uncertainty or probability.
- Requires continuous updates to reflect evolving medical knowledge.
- Cannot replace expert medical judgment in complex cases.

Conclusion

The Smart Medical Diagnosis System successfully applies **Knowledge Representation** and **Logical Reasoning** to perform preliminary patient diagnosis. The medical rules were represented using **Propositional Logic**, and the available facts about Patient A were used to infer possible diseases. Through **Forward Chaining**, the system derived **Possible Flu**and **Possible Pneumonia** as valid diagnoses. Using **Backward Chaining**, the system confirmed that **Patient A has Possible Pneumonia** by proving that both required symptoms**—Cough** and **Breathlessness—**were present. This case study demonstrates how rule-based logical agents can effectively support decision-making in healthcare by providing accurate, explainable, and systematic reasoning based on a predefined knowledge base.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Problem Statement

A smart city traffic controller uses a **First Order Logic (FOL)-based knowledge base** to manage emergency vehicle routing.

Rules

Rule 1

∀x [Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)]

Rule 2

∀x [ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x)]

Rule 3

∀x∀y [Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)]

Facts

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA, Ambulance)

Problem Understanding

An autonomous traffic management system uses Artificial Intelligence to control traffic signals intelligently. The system represents traffic knowledge using **First Order Logic (FOL)** and applies logical inference to provide priority for emergency vehicles such as ambulances.

The objectives of the system are:

- Identify emergency vehicles.
- Automatically clear their route.
- Change traffic signals to green.
- Allow nearby vehicles to move safely.
- Ensure efficient emergency response.

The problem requires the application of **Unification**, **Resolution**, and **Knowledge Representation** to prove logical conclusions.

Analysis of the Problem

The knowledge base consists of three inference rules and four facts.

The inference process is as follows:

Emergency Vehicle

↓

Clear Path

↓

Green Signal

↓

Vehicles behind the ambulance proceed

The system uses logical inference to derive new knowledge from existing facts.

(a) Apply Unification

Definition

Unification is the process of making two logical expressions identical by replacing variables with constants. The resulting substitution is called the **Most General Unifier (MGU).**

Rule 1

$Vehicle(x) \land EmergencyType(x) \rightarrow ClearPath(x)$

Available Facts

Vehicle(Ambulance)

EmergencyType(Ambulance)

Step 1

Match

Vehicle(x)

with

Vehicle(Ambulance)

Substitution

θ1 ={x/Ambulance}

Step 2

Match

EmergencyType(x)

with

EmergencyType(Ambulance)

Substitution

θ2 ={x/Ambulance}

Most General Unifier (MGU)

θ={x/Ambulance}

Apply the Substitution

Rule 1 becomes

Vehicle(Ambulance)

∧

EmergencyType(Ambulance)

→

ClearPath(Ambulance)

Since both facts are present,

Therefore,

ClearPath(Ambulance) is inferred.

Analysis

The variable **x** is successfully replaced by **Ambulance**.

Hence, Rule 1 is activated and the traffic controller concludes that the ambulance should be given a clear path.

(b) Prove Proceed(CarA) using Resolution

Step 1 – Convert Rules into CNF

Rule 1

Vehicle(x)

∧

EmergencyType(x)

→

ClearPath(x)

CNF

¬Vehicle(x)∨¬EmergencyType(x)∨ClearPath(x)

Rule 2

ClearPath(x)

→

GreenSignal(x)

CNF

¬ClearPath(x)∨GreenSignal(x)

Rule 3

Vehicle(x)

∧

Behind(x,y)

∧

GreenSignal(y)

→

Proceed(x)

CNF

¬Vehicle(x)∨¬Behind(x,y)∨¬GreenSignal(y)∨Proceed(x)

Facts in CNF

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA, Ambulance)

Negated Goal

To prove

Proceed(CarA)

Negate the goal

¬Proceed(CarA)

Resolution Process

Resolution 1

Using Rule 1

and

Vehicle(Ambulance)

EmergencyType(Ambulance)

Infer

ClearPath(Ambulance)

Resolution 2

Using Rule 2

ClearPath(Ambulance)

Infer

GreenSignal(Ambulance)

Resolution 3

Apply Rule 3

Substitution

x=CarA y=Ambulance

Rule becomes

Vehicle(CarA)

∧

Behind(CarA,Ambulance)

∧

GreenSignal(Ambulance)

→

Proceed(CarA)

All three premises are true.

Hence,

Proceed(CarA)

is derived.

Resolution 4

Resolve

Proceed(CarA)

with

¬Proceed(CarA)

Result

□

(Empty Clause)

Since an empty clause is obtained,

The negated goal is false.

Therefore,

Proceed(CarA) is TRUE

Resolution Table

Step	Clause	Result
1	Vehicle(Ambulance), EmergencyType(Ambulance)	ClearPath(Ambulance)
2	ClearPath(Ambulance)	GreenSignal(Ambulance)
3	Vehicle(CarA), Behind(CarA, Ambulance), GreenSignal(Ambulance)	Proceed(CarA)
4	Proceed(CarA) + ¬Proceed(CarA)	□ (Empty Clause)

Analysis

Resolution proves the theorem by contradiction.

Since the negated goal leads to an empty clause, the knowledge base logically proves that **CarA is allowed to proceed**.

(c) Analysis of the Knowledge Base

Current Limitation

The present knowledge base only considers one emergency vehicle at a time.

Suppose there are two ambulances approaching the same junction from different directions.

The existing rules cannot determine:

- Which ambulance receives priority.
- Which signal should become green first.
- How conflicting emergency routes are resolved.
- How traffic deadlock can be prevented.

Therefore, the knowledge base is **not sufficient** for multiple simultaneous emergency vehicles.

Additional Rules Required

Rule 4 – Priority Rule

EmergencyPriority(x)>EmergencyPriority(y)→HigherPriority(x)

Purpose:

Assign priority to emergency vehicles.

Rule 5 – Distance Rule

NearIntersection(x)→HigherPriority(x)

Purpose:

Vehicles closer to the junction receive higher priority.

Rule 6 – Conflict Resolution

If two emergency vehicles arrive simultaneously,

then

Select the vehicle with

- shortest distance,
- highest priority,
- earliest arrival time.

Additional Facts

EmergencyPriority(Ambulance1)

EmergencyPriority(FireTruck)

Distance(Ambulance1,30)

Distance(FireTruck,45)

ArrivalTime(Ambulance1,10:15)

ArrivalTime(FireTruck,10:16)

Logical Justification

The additional rules enable the intelligent traffic controller to:

- Resolve conflicts between multiple emergency vehicles.
- Prevent simultaneous green signals.
- Improve traffic flow.
- Reduce emergency response time.
- Ensure road safety.

Without these rules, the system may produce ambiguous decisions or unsafe traffic conditions.

Engineering Interpretation

This case study illustrates how **First Order Logic** can be used to design an intelligent traffic management system. By representing vehicles, emergency conditions, and traffic signals as logical predicates, the system can automatically infer actions such as clearing routes and granting green signals. Unification enables variables to match specific vehicles, while Resolution verifies conclusions through logical proof. However, practical smart-city systems require additional knowledge to handle multiple emergencies, priorities, and dynamic traffic conditions. Extending the knowledge base with priority and conflict-resolution rules makes the system more reliable, scalable, and suitable for real-world deployment.

Critical Analysis

Advantages

- Provides precise logical reasoning.
- Automates emergency vehicle routing.
- Reduces traffic congestion.
- Improves emergency response time.
- Supports explainable AI decision-making.

Limitations

- Assumes complete and correct knowledge.
- Does not account for uncertainty or sensor errors.
- Cannot handle multiple emergency vehicles without additional rules.
- Resolution becomes computationally expensive for very large knowledge bases.
- Requires frequent updates to reflect changing traffic conditions.

Conclusion

The Autonomous Traffic Management Agent successfully demonstrates the application of **First Order Logic, Unification, and Resolution** in intelligent traffic control. Unification produced the Most General Unifier (MGU) **θ = {x/Ambulance}**, allowing Rule 1 to infer **ClearPath(Ambulance)**. Resolution reasoning then proved **Proceed(CarA)** by deriving **GreenSignal(Ambulance)** and resolving the negated goal to the empty clause, confirming the theorem. Although the current knowledge base performs correctly for a single emergency vehicle, it is insufficient for handling multiple simultaneous emergencies. By incorporating additional rules for priority assignment, distance evaluation, and conflict resolution, the system can be extended into a more robust and practical smart-city traffic management solution.

RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand